

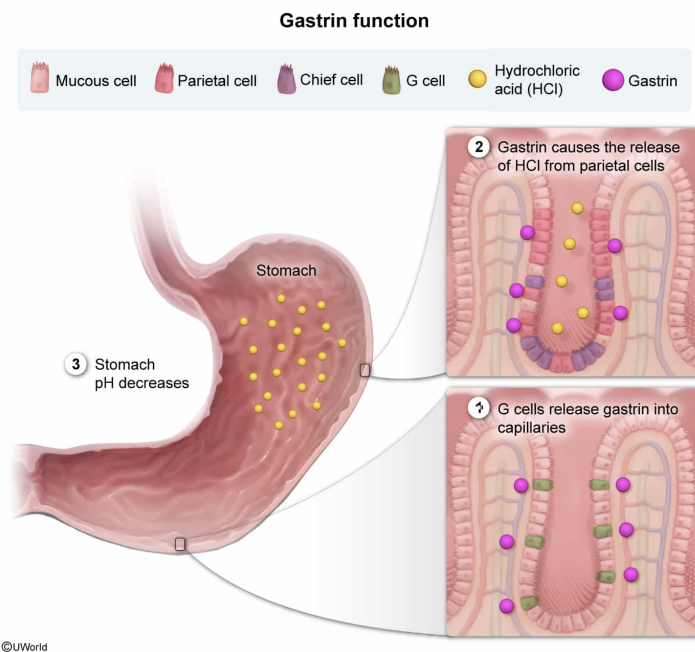
Gastrin regulates the production of hydrochloric acid in the stomach by which type of cells?

- ✔ ☒ A. Parietal cells [51%]
☐ B. Chief cells [18%]
☐ C. Mucous cells [3%]
☐ D. G cells [26%]

Correct

51% answered correctly

Explanation:



As part of the **gastrointestinal tract**, the **stomach** mechanically and chemically digests food, and can also store food before proceeding to subsequent sections of the alimentary canal. **Different cells** of the stomach secrete different products to aid in stomach function:

- **G cells** produce **gastrin**, a hormone that helps regulate secretion of **hydrochloric acid** by parietal cells. The low pH of hydrochloric acid creates an acidic environment in the stomach, which serves to activate proteolytic enzymes, **defend** against pathogens, and denature proteins (**Choice D**).

- **Parietal cells** produce hydrochloric acid and **intrinsic factor**, which aids in the later **absorption** of vitamin B₁₂ in the ileum of the **small intestine**.
- **Chief cells** produce **pepsinogen**, an inactive form (ie, zymogen) of **pepsin**. When activated by the low pH of the stomach, pepsin breaks down polypeptides into smaller peptides. Chief cells also produce **gastric lipase**, which serves to hydrolyze lipids in the stomach **(Choice B)**.
- **Mucous cells** produce **mucus** and **bicarbonate**, which both function to **protect** the walls of the stomach from self-destruction in the stomach's acidic, proteolytic environment **(Choice C)**.

Therefore, **gastrin regulates** the **production of hydrochloric acid** in the stomach by **parietal cells**.

Educational objective:

The stomach secretes gastrin (from G cells), hydrochloric acid and intrinsic factor (from parietal cells), pepsinogen and gastric lipase (from chief cells), and mucus and bicarbonate (from mucous cells). These secretions contribute to the function of the stomach in mechanical and chemical digestion.

Time spent:0 sec

Subject:
Biology

QID:404151

System:
Digestion and Excretion